

Set builder notation

$$\{x \mid x \text{ has property } p\}$$

$$\{1, 2, \dots, 99, 100\} \text{ vs } \{x \mid (x \in \mathbb{Z}) \wedge (1 \leq x \leq 100)\}$$

Equality is evaluated from the top level.

$$\text{Null set} = \emptyset = \{\}$$

Singleton set = set w/ exactly 1 element

Let A, B be sets

If every element $A \in B$

A is a subset of B ,

$$A \subseteq B$$

or B is a superset of A

$$B \supseteq A$$

$$A \subset B$$

$$A \neq B$$

$$A \subseteq B$$

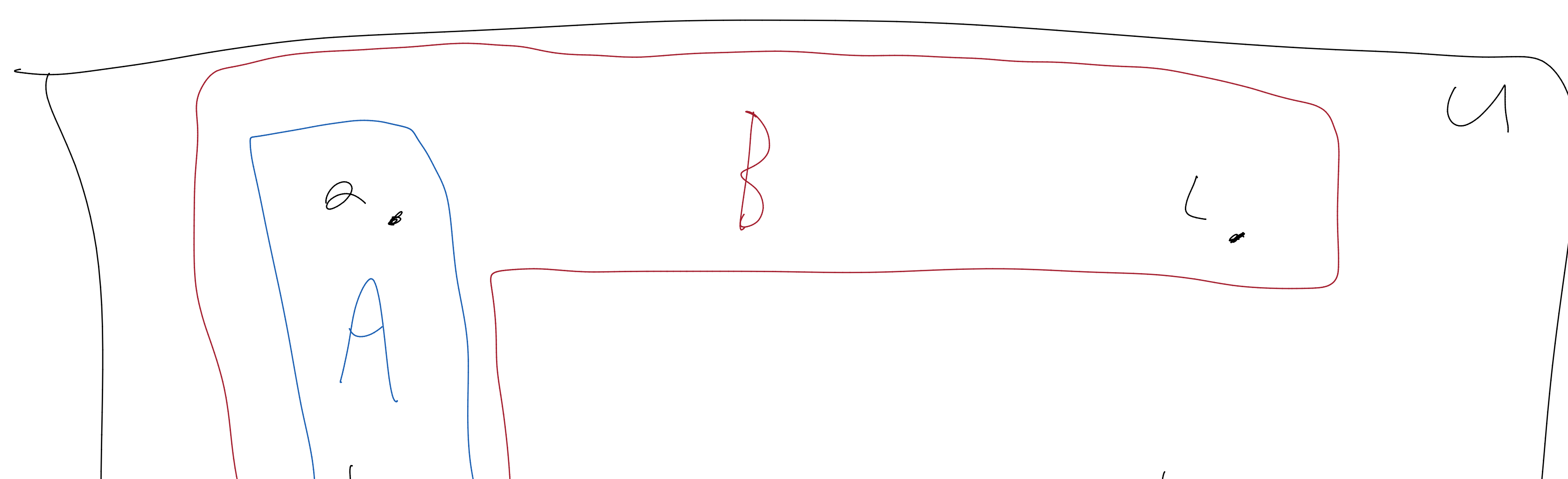
$$A = B$$

Venn diagrams can be used for comparison between sets

Universal set: U is a set that contains all elements, unless otherwise specified.

$$\text{Let } U = \{a, b, c, d\} \quad A = \{a, b\}$$

$$B = \{a, b, c\}$$





$$A \subset B, A \subset U, B \subset U$$

Cardinality means # elements

$$|A| = n, \text{ occasionally } |A| = \infty \text{ e.g. } |\mathbb{Z}| = \infty$$

Power set

$$P(A) = \{B \mid B \subseteq A\}$$

$$|A| = n < \infty \rightarrow |P(A)| = 2^{|A|} = 2^n$$

$$P(\emptyset) = \{\emptyset\}$$

$$P(\{a, b\}) = \{\{a, b\}, \{a\}, \{b\}, \{\emptyset\}\}$$

$$E = \{\emptyset, \emptyset, \{\emptyset, \emptyset\}, \{\emptyset\}, \{\{\emptyset\}\}\}$$

$$|E| = 3$$

2.2

Set operations

Union: $A \cup B$ combine sets and remove dups

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

Intersection: $A \cap B$ elements in both

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

Disjoint: $A \cap B = \emptyset$